

JAVA FOUNDATION ORACLE

1. Which statement is true about the Java Runtime Environment URE

It is a virtual component.

It is a subset of the JVM.

It is platform independent.

It is a subset of the JDK

2. Given the code fragment:

```
int n1 = 10, n2 = 15;
```

```
int sum = 0, avg = 0;
```

```
sum = n1 + n2;
```

```
avg = sum / 2;
```

```
System.out.print("sum = " + sum + " avg = " + avg);
```

What is the result?

A compilation error occurs.

sum = 25 avg = 1

sum = 25 avg = 0

sum = 25 avg = 12

```
Day-9 > J Task.java > Task > main(String[] args)
1  class Task{
    Run main | Debug main
2      public static void main(String[] args) {
3          int n1 = 10, n2 = 15;
4          int sum = 0, avg = 0;
5          sum = n1 + n2;
6          avg = sum / 2;
7          System.out.print("sum = " + sum + " avg = " + avg);
8      }
9  }
10 }
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161> cd Day-9
PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
sum = 25 avg = 12
PS D:\Java_161\Day-9> |
```

3. Given the code fragment:

```
int a = 7;
int b = 3;
int result = a / b + a * b - a % b;
System.out.println (result);
```

What is the result?

22.333334

A runtime exception is thrown.

22

A compilation error occurs.

```
Day-9 > J Task.java > Task > main(String[] args)
1  class Task{
    Run main | Debug main
2      public static void main(String[] args) {
3          int a = 7;
4          int b = 3;
5          int result = a / b + a * b - a % b;
6          System.out.println (result);
7      }
8  }
9  }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
22
PS D:\Java_161\Day-9> 
```

```
4. class Course {

String name;

static int count = 0;

Course(String name) {

this.name = name;

count++;

}

}

public class Sample {

public static void main(String[] args) {

Course c = new Course("Java Programming");

// Line n

}
```

```
}
```

```
System.out.println(name + ":" + count);
```

```
System.out.println(c.name + ":" + count);
```

```
System.out.println(c.name + ":" + Course.count);
```

```
System.out.println(Course.name + ":" + c.count);
```

```
Day-9 > J Sample.java > Sample > main(String[] args)
1  class Course {
2  String name;
3  static int count = 0;
4  Course(String name) {
5  this.name = name;
6  count++;
7  }
8  }
9  public class Sample {
10 public static void main(String[] args) {
11 Course c = new Course("Java Programming");
12 // Line n
13 System.out.println(c.name + ":" + Course.count);
14 }
15 }
16
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Sample.java
PS D:\Java_161\Day-9> java Sample
Java Programming:1
PS D:\Java_161\Day-9>
```

```
5. int a = 10;
```

```
int b = 20;
```

```
int c = 30;
```

```
System.out.println(a++ > 10 || ++b <= 21);
```

```
System.out.println(a > 10 && ++b <= 22);
```

```
System.out.println(a <= 11 && b == 22);
```

```
System.out.println(c++ == 31 && a++ == 11 || b++ == 22);
```

true

true

true

true

false

false

false

false

false

true

false

true

true

false

true

False

```
Day-9 > J Task.java > Task > main(String[] args)
1  class Task{
    Run main | Debug main
2      public static void main(String[] args) {
3          int a = 10;
4          int b = 20;
5          int c = 30;
6          System.out.println(a++ > 10 || ++b <= 21);
7          System.out.println(a > 10 && ++b <= 22);
8          System.out.println(a <= 11 && b == 22);
9          System.out.println(c++ == 31 && a++ == 11 || b++ == 22);
10
11      }
12
13  }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Sample.java
PS D:\Java_161\Day-9> java Sample
Java Programming:1
PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
true
true
true
true
PS D:\Java_161\Day-9> |
```

6. Given the code fragment:

```
List<String> items = new ArrayList<>();
items.add(1, "pen");
items.add(2, "pencil");
items.add(3, "erasers");
items.add("paper");
for (String x : items) {
    System.out.print(x + " ");
}
```

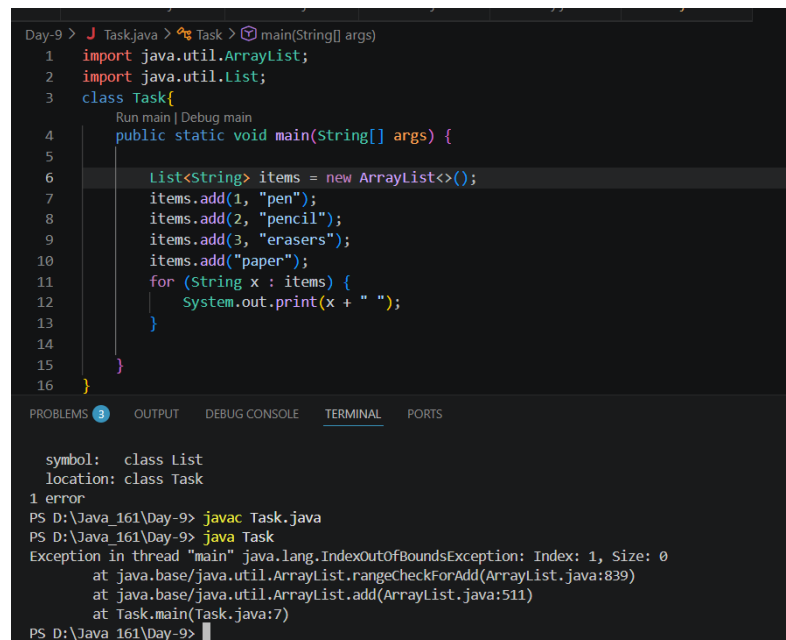
What is the result?

pen pencil erasers paper

A compilation error occurs.

A runtime exception is thrown.

paper pen pencil erasers



```
Day-9 > J Task.java > Task > main(String[] args)
1 import java.util.ArrayList;
2 import java.util.List;
3 class Task{
4     public static void main(String[] args) {
5
6         List<String> items = new ArrayList<>();
7         items.add(1, "pen");
8         items.add(2, "pencil");
9         items.add(3, "erasers");
10        items.add("paper");
11        for (String x : items) {
12            System.out.print(x + " ");
13        }
14    }
15 }
16 }
```

symbol: class List
location: class Task

1 error
PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
Exception in thread "main" java.lang.IndexOutOfBoundsException: Index: 1, Size: 0
at java.base/java.util.ArrayList.rangeCheckForAdd(ArrayList.java:839)
at java.base/java.util.ArrayList.add(ArrayList.java:511)
at Task.main(Task.java:7)
PS D:\Java_161\Day-9>

7. Given the code fragment:

```
int a = 3;
int b = 2;
int c = -1;
if (a + b % c > (a + (-b) * (-c))) {
    System.out.println(a + b - c);
}
```

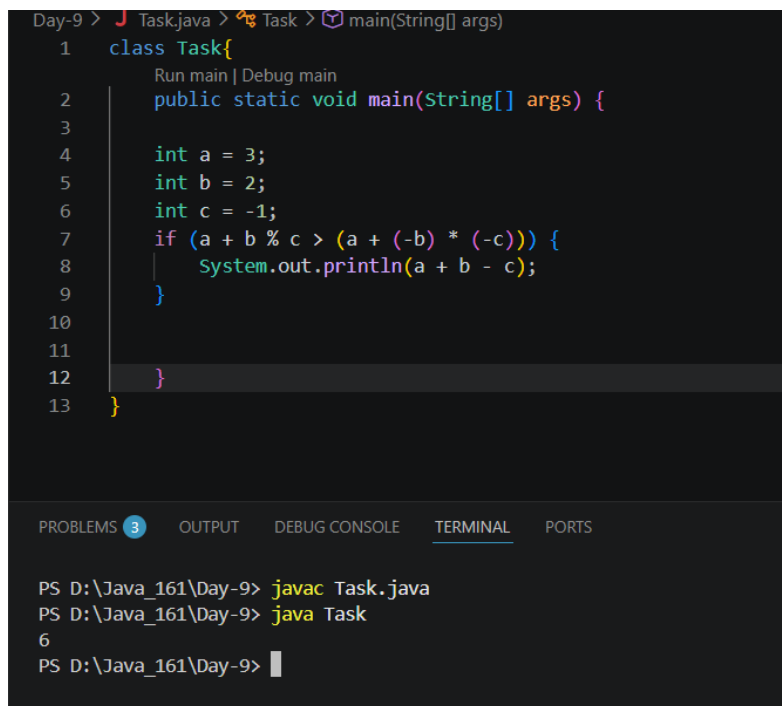
What is the result?

A compilation error occurs.

6

There is no output.

4



The screenshot shows an IDE with a Java file named Task.java. The code defines a class Task with a main method. Inside the main method, three integer variables are declared: a = 3, b = 2, and c = -1. An if statement follows, with the condition (a + b % c > (a + (-b) * (-c))). The body of the if statement contains a single line of code: System.out.println(a + b - c);. The IDE's terminal window at the bottom shows the command prompt with the following sequence of commands and output: PS D:\Java_161\Day-9> javac Task.java, PS D:\Java_161\Day-9> java Task, and the output 6.

```
Day-9 > Task.java > Task > main(String[] args)
1  class Task{
    Run main | Debug main
2      public static void main(String[] args) {
3
4          int a = 3;
5          int b = 2;
6          int c = -1;
7          if (a + b % c > (a + (-b) * (-c))) {
8              System.out.println(a + b - c);
9          }
10
11
12     }
13 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
6
PS D:\Java_161\Day-9> 
```

8. String string1, string2, string3;

string1 = new String("StringObjects");

string2 = string1;

string3 = string1 + string2;

How many String objects are created?

one

two

three

four

```
9 public class Test {  
    public static void main(String[] args) {  
  
        String str1 = "1" + "2";  
  
        System.out.print(str1);  
    }  
}
```

Assume that 1, 2, and 3 are the Unicode values of the characters and respect

What is the result?

3

12

ab

c

10. You have a microprocessor board, such as a Raspberry Pi, wired to control a drone.

Which edition of Java is geared towards use on simple, closed systems with constrained memory requirements, such as a microprocessor board?

Java Enterprise Edition

Java SE Embedded

Java Micro Edition

Java Standard Edition with a Compact Profile

11.

What is the output?

```
String greeting = "Hello";
```

```
boolean equal = (greeting.compareToIgnoreCase("hello") == 0)
? true : false;
```

```
boolean contains = greeting.contains("ello") ? true : false;
```

```
System.out.print(equal + " " + contains);
}
```

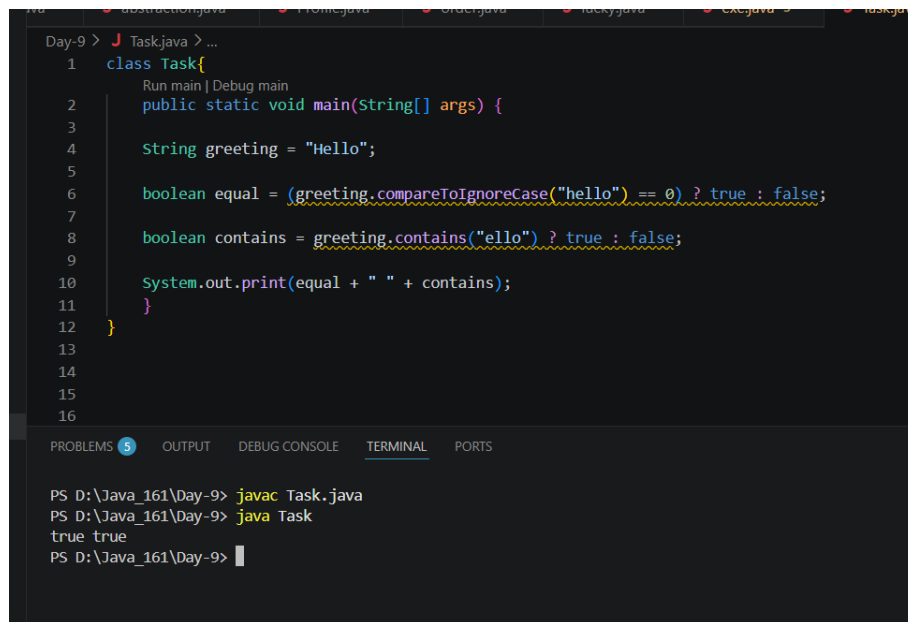
}

Compilation fails.

true false

false false

true true



The screenshot shows an IDE window with a Java file named Task.java. The code defines a class Task with a main method. Inside the main method, a String greeting is set to "Hello". Two boolean variables are calculated: 'equal' is true because "Hello".compareToIgnoreCase("hello") is 0, and 'contains' is true because "Hello".contains("ello") is true. The main method prints the values of 'equal' and 'contains' separated by a space. Below the code editor, the terminal shows the command 'javac Task.java' being executed successfully, followed by 'java Task', which outputs 'true true'.

```
Day-9 > J Task.java > ...
1  class Task{
    Run main | Debug main
2      public static void main(String[] args) {
3
4      String greeting = "Hello";
5
6      boolean equal = (greeting.compareToIgnoreCase("hello") == 0) ? true : false;
7
8      boolean contains = greeting.contains("ello") ? true : false;
9
10     System.out.print(equal + " " + contains);
11 }
12 }
13
14
15
16

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
true true
PS D:\Java_161\Day-9> |
```

12. Which reserved word Controls a java programs execution flow?

While

Private

Short

Class

13. Given the code fragment:

```
}
```

```
// line n1
```

```
switch (var) {
```

```
case "1":
```

```
System.out.println("one");
```

```
break;
```

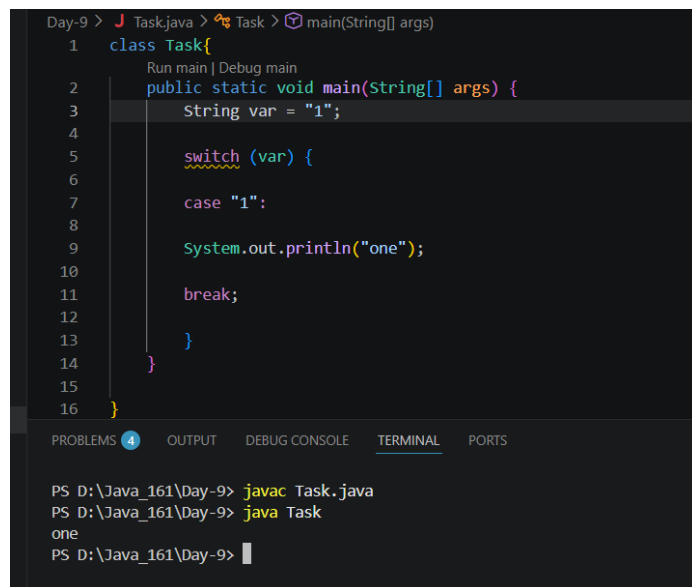
Which code fragment, when inserted at line n1, enables the code to print one

String var = "1";

```
int var= 1;
```

```
char var = '1';
```

```
String var = 1;
```



The screenshot shows an IDE window with a Java file named Task.java. The code is as follows:

```
1 class Task{
2     public static void main(String[] args) {
3         String var = "1";
4
5         switch (var) {
6
7             case "1":
8
9                 System.out.println("one");
10
11                break;
12
13            }
14        }
15    }
16 }
```

The IDE interface includes tabs for PROBLEMS (4), OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the following commands and output:

```
PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
one
PS D:\Java_161\Day-9> |
```

14. Given the code fragment:

```
String str[] = { };
```

```
System.out.println(str.length); // line n1
```

```
System.out.println(str[0]);
```

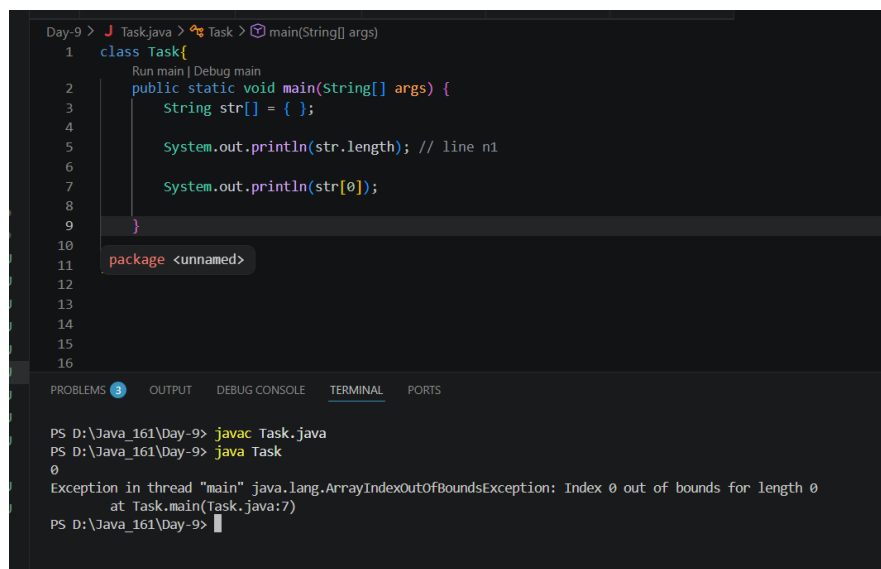
What is the result?

An `ArrayIndexOutOfBoundsException` is thrown at line n1.

0 Then throws `ArrayIndexOutOfBoundsException`

Compilation fails

Throws `NullPointerException` at line n1



```
Day-9 > J Task.java > Task > main(String[] args)
1  class Task{
2      Run main | Debug main
3      public static void main(String[] args) {
4          String str[] = { };
5          System.out.println(str.length); // line n1
6
7          System.out.println(str[0]);
8
9      }
10
11  package <unnamed>
12
13
14
15
16

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Java_161\Day-9> javac Task.java
PS D:\Java_161\Day-9> java Task
0
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 0 out of bounds for length 0
    at Task.main(Task.java:7)
PS D:\Java_161\Day-9>
```

15. Given the code fragment

```
int[] array = new int[5];  
for (int idx =1; idx <= array.length; idx++)  
System.out.print(array[idx]);
```

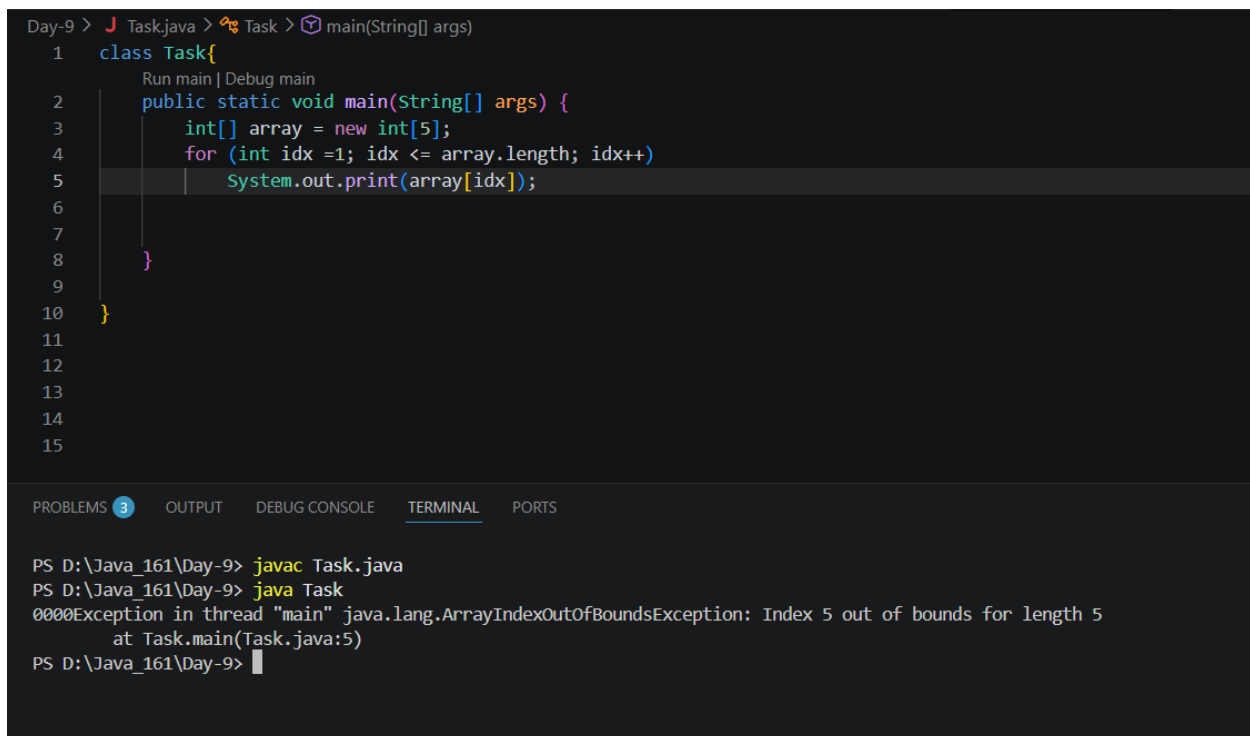
What is the result?

Compilation fails.

Nothing is printed.

00000

0000 followed by an `ArrayIndexOutOfBoundsException` stack trace



```
Day-9 > J Task.java > Task > main(String[] args)  
1 class Task{  
2     Run main | Debug main  
3     public static void main(String[] args) {  
4         int[] array = new int[5];  
5         for (int idx =1; idx <= array.length; idx++)  
6             System.out.print(array[idx]);  
7  
8     }  
9  
10 }  
11  
12  
13  
14  
15  
  
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS  
  
PS D:\Java_161\Day-9> javac Task.java  
PS D:\Java_161\Day-9> java Task  
0000Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 5  
    at Task.main(Task.java:5)  
PS D:\Java_161\Day-9>
```

16. Which code fragment result in a compilation error ?

```
switch (10 < 20){  
case true:
```

```
System.out.println("OK");  
Break;  
}
```

```
for (int c=10; c <20; c++) {  
System.out.println("OK");  
}
```

```
do {  
System.out.println("OK");  
} while (10 < 20);
```

```
if (10 < 20) {  
System.out.println("OK");
```

17. Given:

```
public class Calculator (  
int value;  
public static boolean find() {  
if (value > 100) (  
return true;  
}
```

```
if (value > 50){
```

```
return true;
}
// line n1
}
}
```

Why does the compilation of the Calculator class fail?

It must have an else or else if section at line n1

It accesses non-static variables.

It has multiple return statements

It must have a return statement at line n1.

```
1 public class Calculator {
2     int value;
3     public static boolean find() {
4         if (value > 100) {
5             return true;
6         }
7     }
8     if (value > 50){
9         return true;
10    }
11    // line n1
12    }
13    }
14 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161> cd Day-9
PS D:\Java_161\Day-9> javac Calculator.java
Calculator.java:4: error: non-static variable value cannot be referenced from a static context
    if (value > 100) {
        ^
Calculator.java:8: error: non-static variable value cannot be referenced from a static context
    if (value > 50){
        ^
2 errors
PS D:\Java_161\Day-9>
```

18.the code fragment:

```
public static void main(String[] args) {  
  
    int[] anArray = {1, 3, 5, 7, 11};  
  
    anArray [3]= anArray[1] + anArray[2];  
  
    java.util.Arrays.sort(anArray);  
  
}  
  
for (int x: anArray) (System.out.print(x+" " );}
```

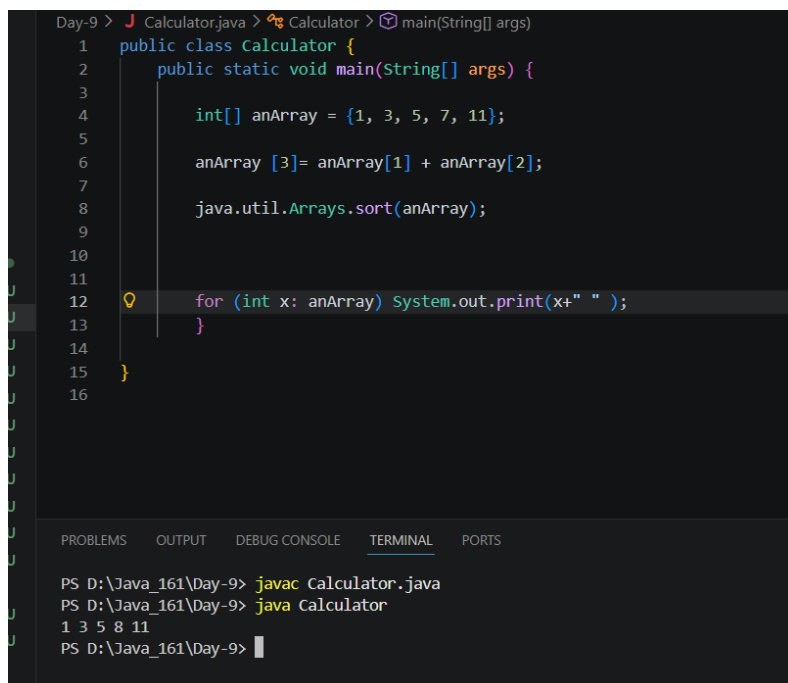
What is the result?

13 4 7 11

1 3 5 8 11

11 74 31

118531



```
Day-9 > J Calculator.java > Calculator > main(String[] args)  
1 public class Calculator {  
2     public static void main(String[] args) {  
3  
4         int[] anArray = {1, 3, 5, 7, 11};  
5  
6         anArray [3]= anArray[1] + anArray[2];  
7  
8         java.util.Arrays.sort(anArray);  
9  
10  
11  
12         for (int x: anArray) System.out.print(x+" " );  
13     }  
14 }  
15  
16  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
  
PS D:\Java_161\Day-9> javac Calculator.java  
PS D:\Java_161\Day-9> java Calculator  
1 3 5 8 11  
PS D:\Java_161\Day-9> 
```


19. Given the code fragment:

```
int num = 0;
```

```
// line n1
```

```
System.out.println(num);
```

Which code fragment can be inserted at line n1 to print a number from 0 to 9?

Which code fragment can be inserted at line n1 to enable the code to print a number from to

```
Random r2 = new Random();
```

```
Num=r2.nextInt(10);
```

```
Random r4= new Random();
```

```
num r4.nextInt() 10;
```

```
numrl.nextInt();
```

```
Random r1 = new Random(10);
```

```
r3.setSeed (10);
```

```
Random r3 = new Random();
```

```
num r3.nextInt();
```

20 . Given the code fragment:

```
public static void main(String[] args) {
```

```
int[] anArray = {1, 3, 5, 7, 11};
```

```
anArray[3] = anArray[1] + anArray[2];
```

```
java.util.Arrays.sort(anArray);
```

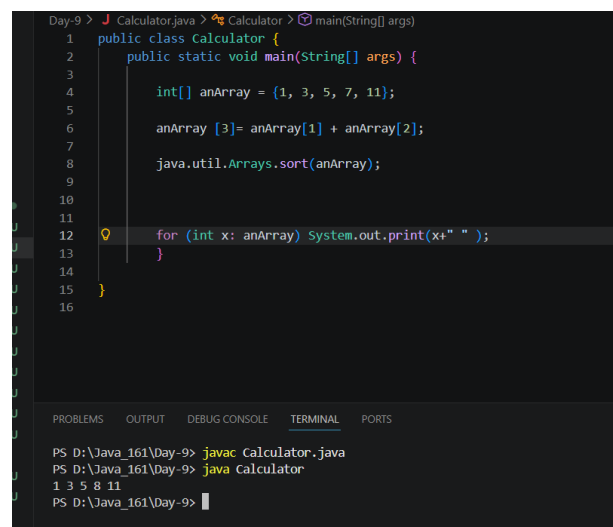
```
for (int x : anArray) {
```

```
System.out.print(x + " ");
```

What is the result?

1 3 4 7 11

1 3 5 8 11



The screenshot shows an IDE with a Java file named 'Calculator.java'. The code in the file is as follows:

```
1 public class Calculator {
2     public static void main(String[] args) {
3
4         int[] anArray = {1, 3, 5, 7, 11};
5
6         anArray [3]= anArray[1] + anArray[2];
7
8         java.util.Arrays.sort(anArray);
9
10
11
12         for (int x: anArray) System.out.print(x+" ");
13     }
14 }
15
16
```

Below the code editor, the 'TERMINAL' tab is active, showing the following commands and output:

```
PS D:\Java_161\Day-9> javac Calculator.java
PS D:\Java_161\Day-9> java Calculator
1 3 5 8 11
PS D:\Java_161\Day-9>
```

11 7431

11 8 5 3 1

Answer: B

21.

```
int x = 10;

int y = 10;

x = x--;

y = --y;

if (x < y) {

x--;

y--;

} else {

x++;

y++;

}

System.out.print("x = " + x + ", y = " + y);

}

}
```

What is the result?

x = 11 y = 11

x = 11 y = 10

x = 10 y = 10

x = 10 y = 11

```
Day-9 > J Calculator.java > Calculator
1 public class Calculator {
2     public static void main(String[] args) {
3
4         int x = 10;
5         int y = 10;
6         x = x--;
7         y = --y;
8         if (x < y) {
9             x--;
10            y--;
11        } else {
12            x++;
13            y++;
14        }
15        System.out.print("x = " + x + ", y = " + y);
16
17    }
18 }
19
20
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Calculator.java
PS D:\Java_161\Day-9> java Calculator
x = 11, y = 10
PS D:\Java_161\Day-9> |
```

22. Which statement is true about arrays and ArrayList instances?

Duplicate values are not allowed in either arrays or ArrayList instances

You can iterate through an array or ArrayList instance in only one direction.

ArrayList instances are resizable whereas arrays are not.

Arrays use a zero-based index whereas ArrayList instances use a one-based index.

23. Given the code fragment:

```
public static void main(String[] args) |  
double num = -25.67;  
System.out.println(Math.abs(num));
```

What is the result?

26

25.7

25.00

25.67

```
Day-9 > J Calculator.java > Calculator
1 public class Calculator {
2     public static void main(String[] args) {
3
4         double num = -25.67;
5         System.out.println(Math.abs(num));
6
7     }
8 }
9
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Calculator.java
PS D:\Java_161\Day-9> java Calculator
25.67
PS D:\Java_161\Day-9> 
```

24. Given the Classes:

Apple.java

package fruits;

```
class Apple {
    public void getApple() {
        System.out.println("Apple method");
    }
}
```

Salad.java

package food;

// line n1

```
class Salad {  
  
    public void prepareSalad() {  
        Apple apple = new Apple(); // line n2  
        apple.getApple();  
    }  
}
```

Which two modifications, independently, enable the `Salad.java` file to compile?

Add `import fruits.Apple;` at line n1

Change class `Apple` to `public class Apple`

Move `Salad` class to `fruits` package

Make `getApple()` method `private`

Remove package declaration from `Apple.java`

Answer: B, C

25. Which class definition is valid?

```
class A {  
  
    public static void methodA(String[] args) {
```

```
}  
}
```

```
A class {  
void methodA(String[] args) ()
```

```
private class A {  
public void methodA(String args) ( }  
}
```

```
class A {  
public void static methodA(String[] args) ( )  
}
```

26. Given the Car. java file:

```
public class Car {  
public static void main(String[] args) |  
System.out.print(args[0]);
```



```
}avac
```

Which option enables you to print win !?

```
javac Car.java
```

```
java Car Win!
```

```
javac Car.java Win!
```

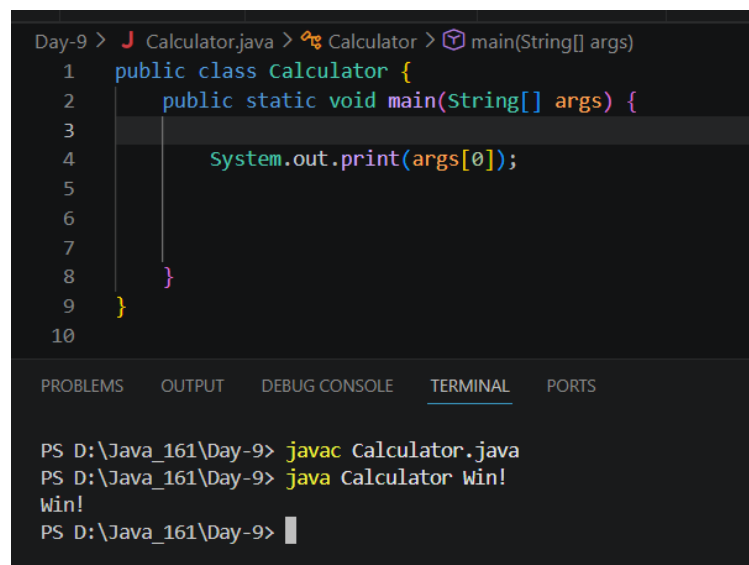
```
java Car.class
```

```
javac Car.java
```

```
java Car.class Win!
```

```
javac Car.java "Win!"
```

```
java Car
```



The screenshot shows an IDE with a Java file named 'Calculator.java' open. The code defines a 'Calculator' class with a 'main' method that prints the first command-line argument. The terminal output shows the compilation and execution of the program, resulting in 'Win!' being printed.

```
Day-9 > J Calculator.java > Calculator > main(String[] args)
1 public class Calculator {
2     public static void main(String[] args) {
3
4         System.out.print(args[0]);
5
6
7     }
8 }
9
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-9> javac Calculator.java
PS D:\Java_161\Day-9> java Calculator Win!
Win!
PS D:\Java_161\Day-9> 
```

Given:

```
class Ball {  
    boolean usable;  
  
    Ball ball;  
  
    public static void main (String[] args) |  
  
        Ball b = new Ball();  
  
        System.out.print(b.usable);  
  
        System.out.print(" "+ b.ball.usable);  
  
    }  
  
}
```

What is the result?

false

A runtime exception is thrown.

true

A runtime exception is thrown.

true true

false false

```
Day-9 > J Ball.java > Ball > main(String[] args)
1 class Ball {
2     boolean usable;
3     Ball ball;
4     Run main | Debug main
5     public static void main (String[] args) {
6         Ball b = new Ball();
7         System.out.print(b.usable);
8         System.out.print(" " + b.ball.usable);
9     }
10 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Java_161\Day-9> javac Ball.java
PS D:\Java_161\Day-9> java Ball
falseException in thread "main" java.lang.NullPointerException: Cannot read field "usable" because "<local1>.ball" is null
    at Ball.main(Ball.java:7)
PS D:\Java_161\Day-9> 
```

